Joohan Bae

M.Eng. Student, Department of Mechanical Engineering, Kogakuin University, Tokyo, Japan
am26074@g.kogakuin.jp · joohanbae.github.io · ORCID 0009-0001-2834-5196

RESEARCH INTERESTS

Human–AI interaction; interaction techniques that help users and generative AI align intent and interpretation; visualizing how AI interprets instructions for non-expert users; human–robot interaction

EDUCATION

Kogakuin University, Tokyo, Japan

M.Eng. in Mechanical Engineering — Apr 2026 – Present

- Advisor: Prof. Daigo Misaki (Misaki Laboratory)
- Early completion under faculty review; expected Mar 2027 if approved
- GPA: 3.79/4.0 (16 credits completed as of Oct 2026)

B.Eng. in Mechanical Systems Engineering — Apr 2020 – Mar 2026

- GPA: 3.6/4.0
- Leave of absence for mandatory military service, Republic of Korea (Apr 2022 – Mar 2024)

PUBLICATIONS

Peer-reviewed posters

1. **Joohan Bae** and Daigo Misaki. 2026. Who Pays for Vagueness? A Visual Token-Metered Interaction Technique for Promoting Precise AI Commands with TokenTangram. In *Adjunct Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology (UIST '26 Adjunct)*. Poster. To appear. doi:10.1145/3830397.3841836
2. Yuya Aikawa, **Joohan Bae**, and Daigo Misaki. 2025. Clay Play: A Metaphorical LLM Interface for Dynamic Idea Generation and Exploration. In *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25 Adjunct)*, Busan, Republic of Korea. Poster. doi:10.1145/3746058.3758345

Under review and forthcoming

3. Full paper on how users form and revise capability beliefs about an LLM-enabled tabletop robot (second author). *ACM/IEEE International Conference on Human-Robot Interaction (HRI '27)*. Under review; details withheld during double-blind review.
4. **Joohan Bae**, Victor Taratukhin, and Daigo Misaki. Converging Through Speech or Through AI-Refined Text: Four Team Cases from a Hands-On Design Thinking Seminar for Engineering Students. *Journal of Robotics and Mechatronics*. Under revision.
5. Daigo Misaki, **Joohan Bae**, and Komei Miura. Designing Human–AI Interaction to Foster Ambidextrous Thinking and Learner Ownership in Product Ideation. *AHFE 2026 Hawaii International Conference: Human Factors in Design, Engineering, and Computing*, Honolulu, HI, USA, Dec 2026. Abstract accepted; full paper to appear.

PRESENTATIONS

- Daigo Misaki and **Joohan Bae**. 2026. Integrating Vibe Coding into Mechanical Systems Engineering Education: From Mechatronics Experiments to Prototyping in Undergraduate Thesis Research. 25th International Symposium on Advanced Technology (ISAT-25), Southern Taiwan University of Science and Technology, Tainan, Taiwan, Oct 2026. Poster.
- Nana Sakai, Mayu Ikeda, **Joohan Bae**, Genki Sakai, Daigo Misaki, and Victor Taratukhin (equal contribution). 2026. Prototype of a Conversational Robot with Facial Expressions and Body Movements for Sustained Dialogue with Older Adults (in Japanese). 12th Medicine–Pharmacy–Engineering Three-University Symposium, Tokyo, Japan, Oct 2026. Poster.

RESEARCH EXPERIENCE

Master's Research, Misaki Laboratory, Kogakuin University Apr 2026 – Present

Advisor: Prof. Daigo Misaki

- Developing interaction techniques that make the process of aligning intent and interpretation between users and generative AI visible, so users can understand and correct AI outputs themselves (thesis: *Interaction Support for Meaning Negotiation in Human–Generative AI Collaboration*)
- Defined implicit assumptions as what is unstated yet necessary for an AI result, and operationalized them as measurable interaction traces
- Designed and built TokenTangram, which meters token counts per command and cumulatively as a processing-load indicator and replays how the AI interpreted each command (UIST '26 poster, first author)
- Designing a 12-participant within-subjects study comparing length-based and attempt-based cost displays (Dec 2026)

Kogakuin University × University of Münster Joint Project-Based Learning 2026

- Led the Japanese team's functional development (circuits, firmware, interaction design) of IkigAI, a communication robot for older adults, in a 15-member team (11 German, 4 Japanese)
- Coordinated communication with the German team in English through in-person and online meetings
- Built an LLM-driven behavior system on Raspberry Pi 5 and M5Stack CoreS3 that maps emotional responses to servo motions
- Diagnosed hardware failures, including SPI signal-integrity issues and audio amplifier damage from servo back-EMF
- Conducted a 10-participant human–robot interaction study with the robot

Design Thinking Workshop Study, Misaki Laboratory

- Compared how four teams (16 engineering students) converged on an idea in a 4.5-hour design-thinking workshop, through spoken discussion or AI-refined text
- Analyzed convergence ratings, transcripts, and AI chat logs; first-authored the journal manuscript

TEACHING EXPERIENCE

Teaching Assistant, Design Engineering, Kogakuin University Sep 2026 – Present

SCHOLARSHIPS

- Kogakuin University Scholarship for Privately Financed International Students with Outstanding Academic Performance — 2025, 2026
- Taisei Oncho Scholarship Foundation Scholarship (two-year award) — 2025 – 2026
- Monbukagakusho Honors Scholarship for Privately-Financed International Students, Japan Student Services Organization (JASSO) — 2021, 2024

LEADERSHIP AND ACTIVITIES

Kogakuin Robot Project (KRP) 2020 – 2022

- Led the programming team (2021 – 2022), directing development of robot control software
- Developed robot control programs as a programming team member (2020 – 2021)

SKILLS

- **Programming:** Python, C, JavaScript
- **Hardware prototyping:** Raspberry Pi, Arduino, M5Stack; sensor and actuator control with basic circuits
- **CAD:** Autodesk Fusion
- **Research methods:** User studies, questionnaires, and interviews; statistical analysis of questionnaire data (ANOVA, MANOVA); MATLAB

LANGUAGES

Korean (native) · Japanese (fluent) · English (professional working proficiency)

Last updated October 2026